

BattleBuddy: An AI Machine Learning Companion

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Executive Summary

The BattleBuddy platform is an AI-powered educational web platform developed to support veterans and non-traditional students pursuing careers in technology. The platform provides adaptive quizzes, an intelligent chatbot, and curated learning materials aligned with accelerated certification and professional development programs.

Designed to serve prospective, current, and graduated students, the platform features role-based access, topic-driven learning paths, and dynamic progress tracking. This project is being developed using Django and Python for the backend, with a modular architecture to support future scalability and cloud deployment. Testing and development are currently being conducted on a local server environment as part of the project's proof of concept phase.

WAV2T aims to address institutional goals by improving student onboarding, success, and certification readiness while enabling data-driven decision-making by administrators. The system's design prioritizes accessibility, security, and usability, ensuring it can grow alongside the needs of the program and its stakeholders.

I. Introduction

1. Problem Statement

Many veterans and non-traditional students face barriers when transitioning into academic programs and technology careers. These barriers include gaps in foundational knowledge, lack of preparation for entrance exams, and difficulty accessing personalized study resources. Additionally, program administrators face challenges in assessing student readiness, tracking progress, and providing targeted support.

There is a clear need for an integrated, adaptive, and user-friendly platform that guides students through academic preparation while offering actionable insights to instructors and staff.

2. Project Objectives

The WAV2T project seeks to develop a secure, accessible, and scalable web-based platform that addresses the following key objectives:

- Provide prospective students with entrance and skills assessment exams to determine readiness for program pathways.
- Offer adaptive practice exams and study materials that adjust to individual student performance.
- Support current and graduated students preparing for industry-recognized certifications.
- Deliver actionable insights and data analytics to administrators for tracking program outcomes and continuous improvement.
- Offer on-demand study support through a chatbot powered by OpenAI.
- Ensure compliance with data privacy, accessibility, and institutional IT standards.

3. Project Scope

The WAV2T platform includes the following core features:

- **Secure Login & Role-Based Access:** Supports prospective, current, and graduated students as well as administrators.
- **Entrance Exams:** Timed, randomized exams that determine placement into one of three learning tracks: Server and Cloud Applications (SCA), Cloud Application Development (CAD), or Cybersecurity Administration.

- **Adaptive Practice Exams:** Real-time difficulty adjustment based on student performance using backend logic tied to student progress.
- **Study Materials Repository:** Pathway-specific content including videos, documents, and practice exams.
- **AI-Powered Chatbot:** An intelligent assistant that answers study-related questions and helps users navigate the platform.
- **Progress Tracking & Reporting:** Allows students to monitor their improvement while giving admins insight into student performance trends.

The project is currently in a proof-of-concept phase with local hosting, using Django for the backend and static HTML templates for the front end. Full deployment, including cloud hosting, email integration, and advanced analytics, is planned for future phases.

4. Stakeholders

The following stakeholders are directly impacted by the WAV2T platform:

- **Prospective Students:** Individuals evaluating their readiness for entry into one of the WAV2T educational pathways.
- **Current Students:** Enrolled participants using the platform to access study materials and practice exams.
- **Graduated Students:** Alumni seeking to prepare for industry certifications through continued access to learning tools.
- **Administrators and Faculty:** Responsible for managing exams, monitoring student progress, and using analytics for program decisions.
- **Engineering Advisory Board (EAB):** Provides feedback and oversight on platform functionality and strategic alignment.
- **Development Team:** Student developers responsible for designing, building, and maintaining the platform during its initial phases.
- **Dr. Dvorak:** Faculty sponsor ensuring academic oversight and alignment with educational outcomes.

II. Background and Literature Review

1. Background

As the demand for skilled technology professionals grows, educational institutions are under increasing pressure to create accessible and effective pathways into technical careers. This challenge is especially critical for veterans and non-traditional students, who often bring valuable real-world experience but may lack recent academic exposure or familiarity with modern learning platforms.

Traditional educational systems frequently fail to provide personalized, adaptable resources to support these learners, leading to high attrition rates and underutilized talent. The Washington Vets 2 Tech (WAV2T) initiative was established to bridge this gap by offering a structured, technology-driven learning platform that caters specifically to the unique needs of transitioning service members and non-traditional learners.

WAV2T's goal is not only to support academic entry through entrance and skill assessment exams but to foster long-term career readiness by offering adaptive study tools and certification preparation. The platform is designed with flexibility in mind, enabling use by prospective, current, and graduated students, each with distinct learning needs and goals.

2. Literature Review

2.1 Adaptive Learning Systems

The concept of adaptive learning has gained significant traction in recent years. According to research by Brusilovsky and Millán (2007), adaptive learning systems personalize educational content by tracking user performance and adjusting instruction accordingly. These systems use learner models to tailor difficulty levels, content sequencing, and feedback in real time.

WAV2T incorporates this model by tracking student quiz performance using Django-based progress models. The backend dynamically adjusts the difficulty of subsequent questions, creating a learning experience that evolves with the student—much like an intelligent tutor. This personalized approach improves learner engagement and outcomes by meeting students where they are academically.

2.2 Technology for Veteran and Non-Traditional Learners

Multiple studies (e.g., Griffin & Gilbert, 2015; Vacchi & Berger, 2014) emphasize the unique needs of veterans in higher education. These include the need for flexible learning, technology-supported study aids, and systems that validate prior experience while building new skills. WAV2T is directly aligned with these findings, offering self-paced entrance assessments and adaptive materials designed to support knowledge refreshment and skill-building.

The system's structure supports asynchronous learning and personalized tracking, reducing stress and offering a smoother transition for users who may be re-entering formal education after a gap.

2.3 Chatbots and AI in Education

AI-powered chatbots have become increasingly prevalent in higher education. Studies like Winkler & Söllner (2018) show that well-designed educational chatbots can enhance learner satisfaction, increase help-seeking behavior, and reduce instructor burden. WAV2T integrates OpenAI's language model to create a chatbot study buddy that supports student navigation, answers course-related questions and improves engagement through immediate feedback and guidance.

This aligns with ongoing research that suggests conversational agents can significantly improve accessibility, especially for students who may hesitate to seek help from instructors.

2.4 Technical Foundation: Web-Based Educational Platforms

The technical foundation of WAV2T is grounded in modern web development best practices. Built using Django (a high-level Python framework), the platform adopts the Model-View-Template (MVT) pattern, which enhances maintainability, scalability, and security. Django's built-in authentication, ORM, and admin tools streamline backend development while ensuring proper data handling and role-based access.

For frontend responsiveness and accessibility, standard technologies such as HTML5, CSS3, JavaScript, and Bootstrap are employed. This ensures that WAV2T is fully compatible with a range of devices and meets accessibility standards such as WCAG 2.1.

Although currently hosted locally during its proof-of-concept stage, the platform's modular design enables seamless migration to scalable cloud solutions such as AWS, Azure, or institutional servers in the future.

3. Relevance and Contribution

WAV2T contributes to the growing field of educational technology by:

- Offering a veteran-centric, modular learning platform tailored to underserved student populations.
- Implementing a data-driven adaptive quiz engine based on proven models of personalized education.
- Integrating AI-driven support tools to improve usability and engagement.

- Aligning with industry certification standards, helping students transition directly into workforce opportunities.

Through the combination of educational theory, technical precision, and accessibility-focused design, WAV2T fills a critical gap in postsecondary and workforce education systems.

III. System Requirements

The WAV2T (Washington Vets 2 Tech) platform is an educational web application developed to support prospective, current, and graduated students in their academic and professional advancement. The system is built with a modular, secure architecture using Python and the Django framework, hosted locally for its proof-of-concept phase with future scalability in mind. Below is an overview of the system requirements, categorized into functional, non-functional, and design constraints.

Functional Requirements

At its core, the WAV2T platform delivers a set of critical features that directly support users along three primary educational pathways: Server and Cloud Applications (SCA), Cloud Application Development (CAD), and Cybersecurity Administration (Cyber). While we understand that the CAD program is not currently being offered, we wanted to showcase the toggle option. This option allows the admin to hide non-active courses allowing for greater flexibility to the dynamic needs of the program. The system allows users to register and authenticate securely using a role-based access model that differentiates between prospective students, enrolled students, graduates, and administrators.

Key functional capabilities include a vast exam management module, enabling administrators to create and administer randomized, timed entrance exams tailored to each learning path. A skills assessment feature further refines placement and study planning. Additionally, the platform offers an extensive study material repository with curated resources, such as videos, texts, and practice exams, and supports an adaptive learning engine. This engine uses machine learning to tailor exam content based on user performance, promoting more personalized and effective study experiences.

Students interact with a streamlined, multi-tabbed user interface that organizes features by educational track. Administrators, in turn, access a centralized dashboard for user management, content updates, exam grading, and performance analytics. Notifications, currently limited to internal push alerts, keep both students and staff informed of grading events, new resources, and progress milestones.

A key innovation within WAV2T is its AI-powered chatbot study assistant, which supports students by answering queries and directing them to appropriate study materials. Though currently focused on student interaction, plans for expanded administrator functionality are included in the system roadmap.

Non-Functional Requirements

To ensure a high-quality user experience, the WAV2T platform incorporates numerous non-functional considerations. The system's user interface is designed for accessibility and responsiveness, complying with Web Content Accessibility Guidelines (WCAG) and optimized for use across a range of devices and screen sizes. The platform prioritizes usability, with the goal that 90% of new users should be able to navigate its core functions without external guidance.

Performance benchmarks include a sub-500ms response time for page loads under typical operating conditions and the ability to support 500 concurrent users without notable degradation in performance. Real-time interactions, such as exam submissions and internal notifications, are processed within a two-second window, ensuring responsiveness critical to timed assessments and progress tracking.

Security is a central concern. Django's built-in authentication mechanisms, combined with role-based access control, password hashing, and session management, provide a strong foundation for user data protection. Future enhancements include two-factor authentication and encryption at rest using AES-256 once cloud deployment begins. The platform is designed to meet FERPA compliance standards and will undergo regular security audits and penetration testing when moved to production.

Maintainability and scalability are also integral to the system's design. Clear code documentation, adherence to PEP8 coding standards, and modular architecture support long-term development. The backend is structured to facilitate deployment via containers or cloud infrastructure such as AWS, with current use of SQLite allowing for rapid prototyping and future database migration paths already outlined.

Constraints

Several constraints have influenced the system's development. The platform is currently hosted locally, limiting real-time scalability and third-party integrations such as email notifications or institutional APIs. SQLite serves as the initial database engine, which, while effective for development, may not scale well with production-level traffic or data volumes. Budget constraints and institutional requirements have also delayed implementation of features such as full email integration, external identity provider support, and advanced reporting dashboards.

Technology choices, including Django as the backend framework, have shaped compatibility and development workflows, aligning the system with open-source ecosystems rather than proprietary tools like .NET. Additionally, accessibility, usability, and security standards impose strict guidelines for interface design, encryption protocols, and user role management, which must be carefully balanced against development timelines and resource limitations.

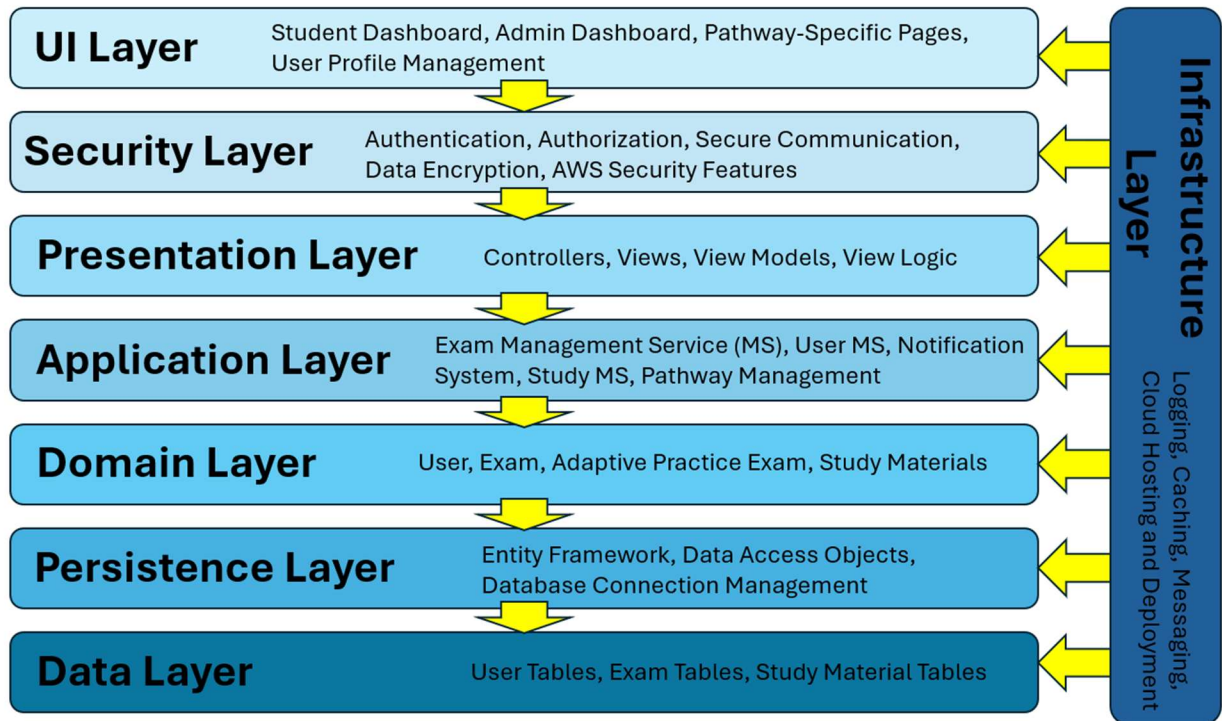
IV. System Design

The system design of the WAV2T platform encompasses the overall architecture, user interface layout, security structure, and major software components. The platform follows a layered, modular architecture that ensures scalability, maintainability, and security. The system is currently in a locally hosted proof-of-concept phase, with infrastructure and design choices that support a seamless transition to full-scale deployment.

1. System Architecture

The WAV2T platform uses a multi-layered architecture composed of the following primary layers:

- **User Interface (Presentation Layer):** Built using Django templates, HTML, CSS, and JavaScript, this layer handles all user interactions and provides an accessible and responsive front-end experience across devices.
- **Application Layer:** Manages the business logic and routes user input to appropriate services. This includes quiz logic, adaptive test mechanisms, chatbot interactions, and dashboard updates.
- **Domain Layer:** Defines the core models and entities (e.g., User, Exam, StudyMaterial) and enforces rules on how these interact.
- **Persistence Layer:** Interfaces with the SQLite database through Django's Object-Relational Mapping (ORM), managing CRUD operations and schema migrations.
- **Security Layer:** Embedded throughout the stack, responsible for authentication, access control, session management, and data protection.

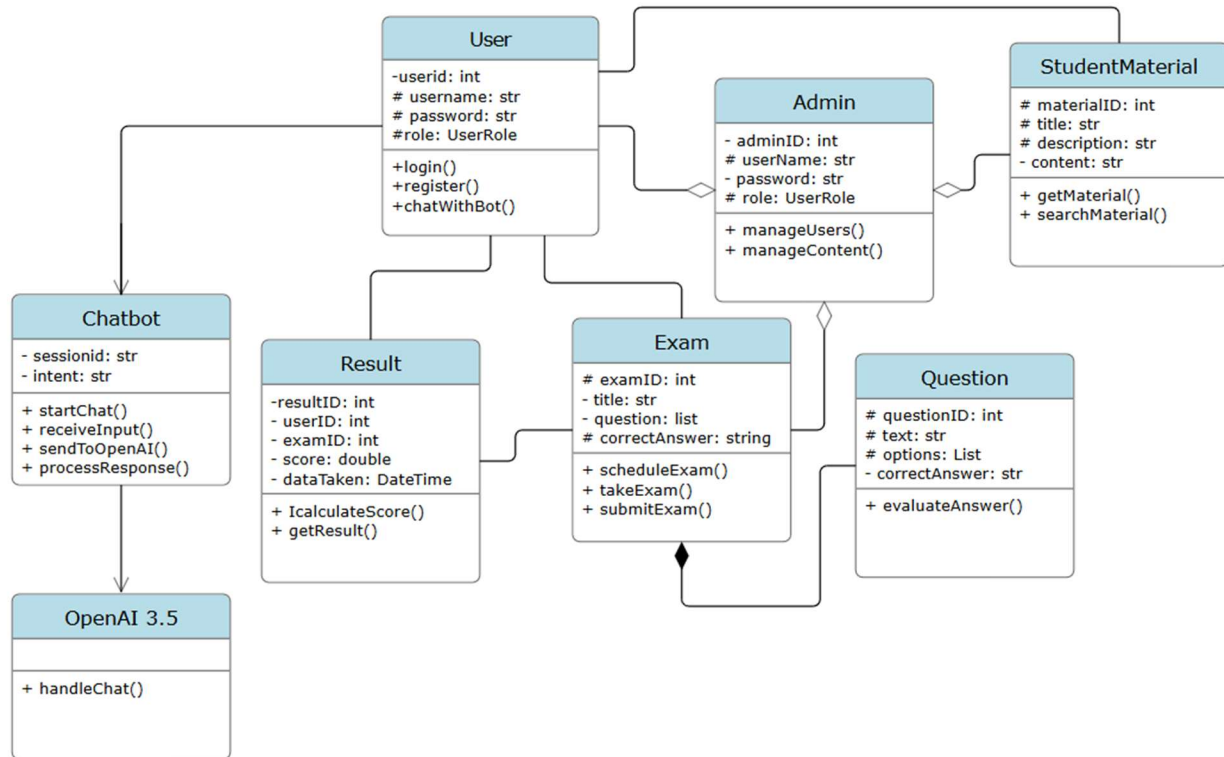


2. UML Design

The platform's main entities are structured using Django models and follow object-oriented design principles. The UML class diagram illustrates the relationships between core objects such as User, Exam, Result, and StudyMaterial.

Key design relationships include:

- One-to-many relationship between User to Exam, User to Result, Admin to User, Admin to Exam, Exam to Result, and Exam to Question.
- One-way relationship between User to Chatbot, and Chatbot to OpenAI.
- Many-to-many relationships between Admin to StudyMaterial.



3. User Interface (UI) Design

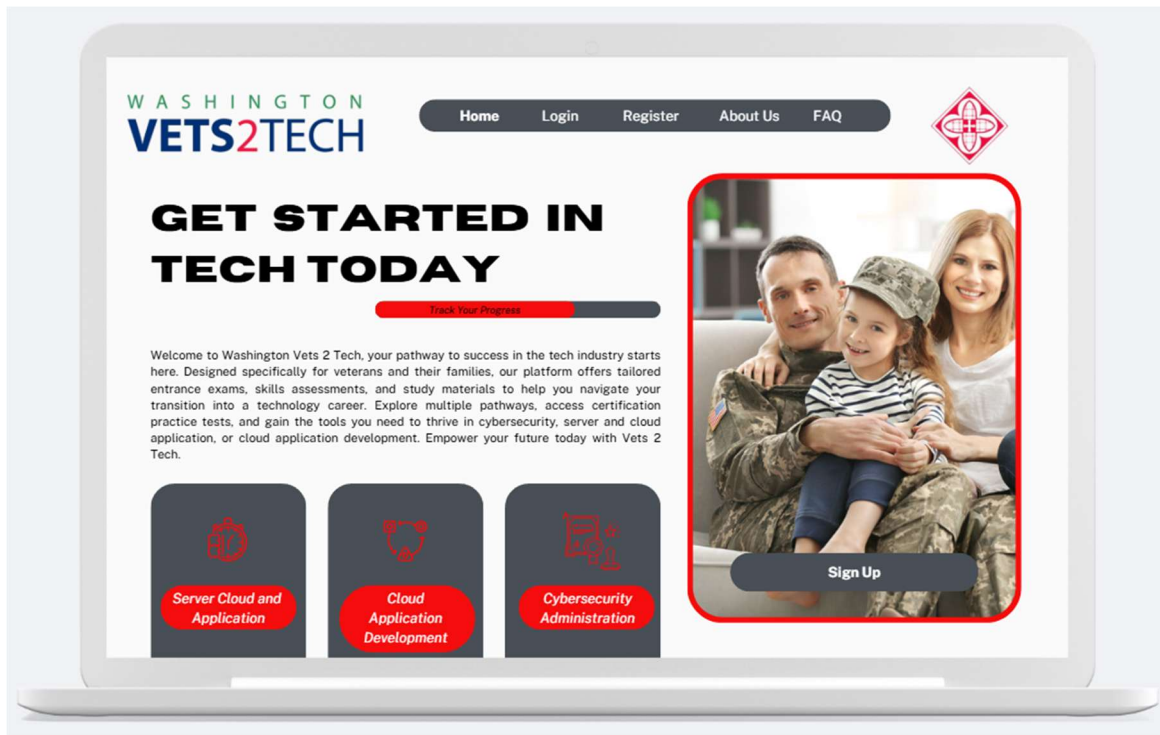
The UI was designed with simplicity and accessibility in mind, ensuring that all users, from prospective students to administrators, can intuitively navigate the platform. The multi-tabbed layout provides clear pathways for SCA, CAD, and Cyber users, while admin users have access to a separate dashboard for content and account management.

Core interface components include:

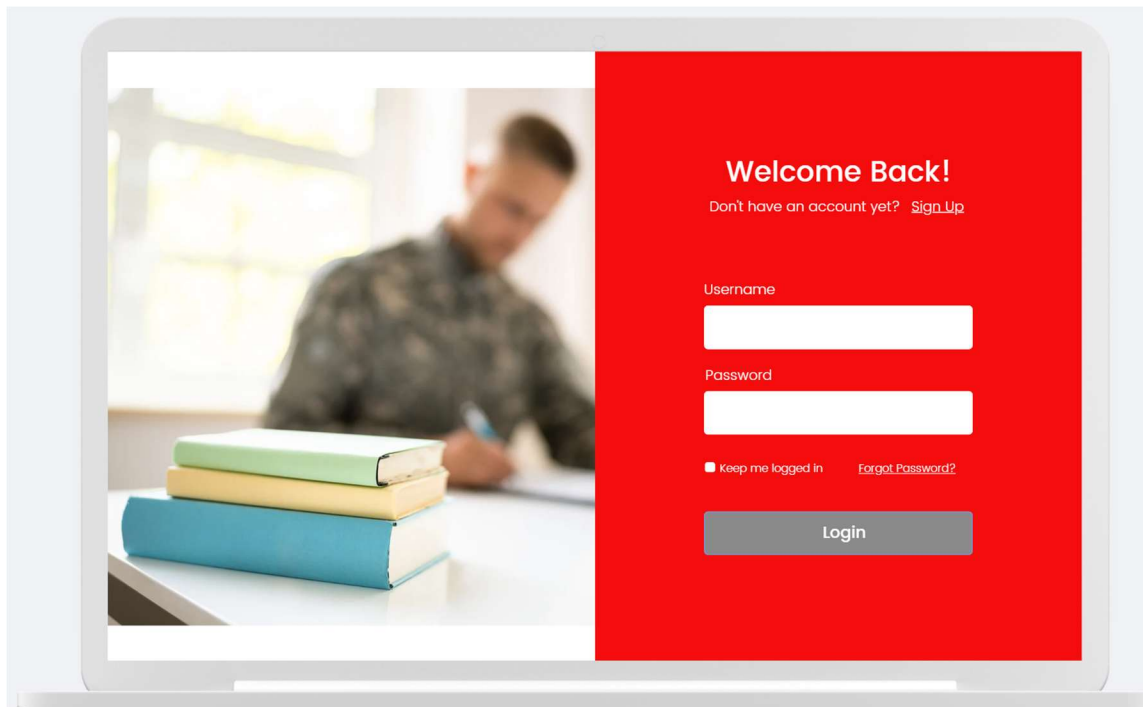
- **Login/Registration Page:** Secure access point with CSRF protection and future-ready two factor authentication (2FA) prompts
- **Student Dashboard:** Personalized with pathway-specific tabs, progress tracking, and study resources
- **Admin Dashboard:** Central hub for exam management, notifications, and analytics
- **Chatbot Interface:** Integrated into all pages, providing contextual support and study guidance

Wireframes:

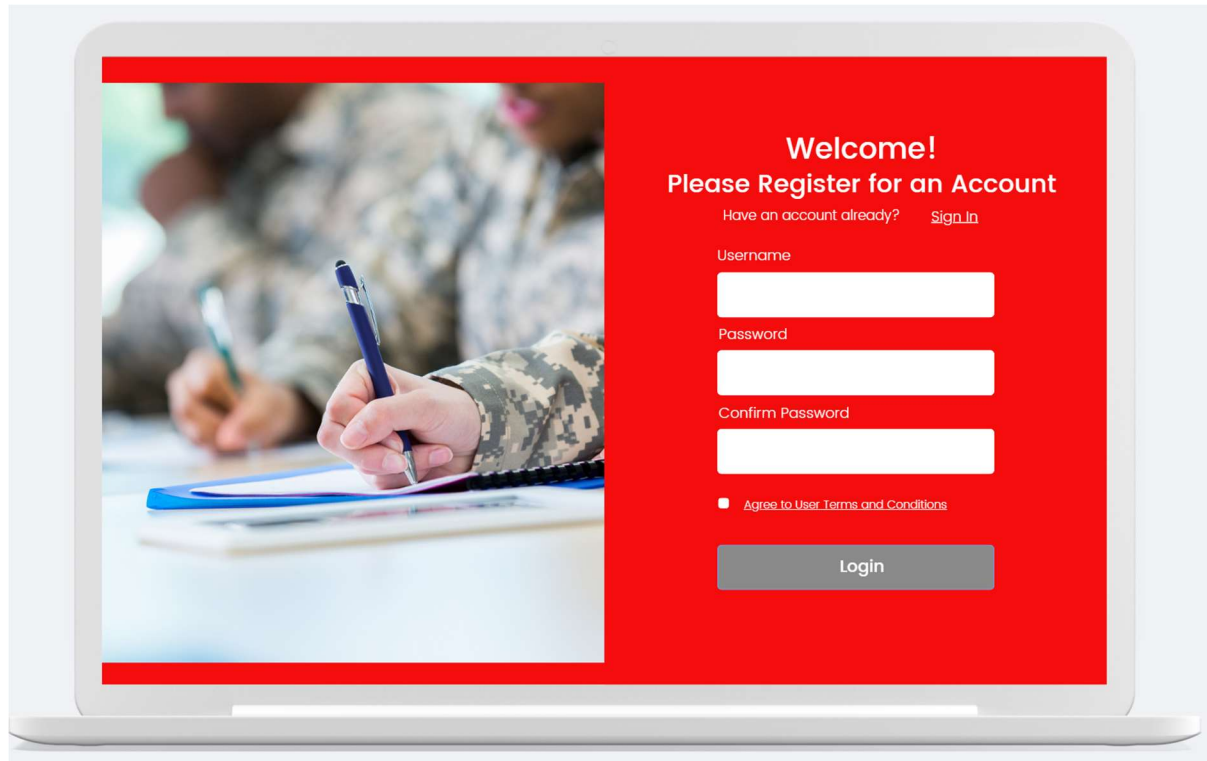
[Figure 1: Home Page]



[Figure 2: Login]



[Figure 3: Register]



Welcome!
Please Register for an Account

Have an account already? [Sign In](#)

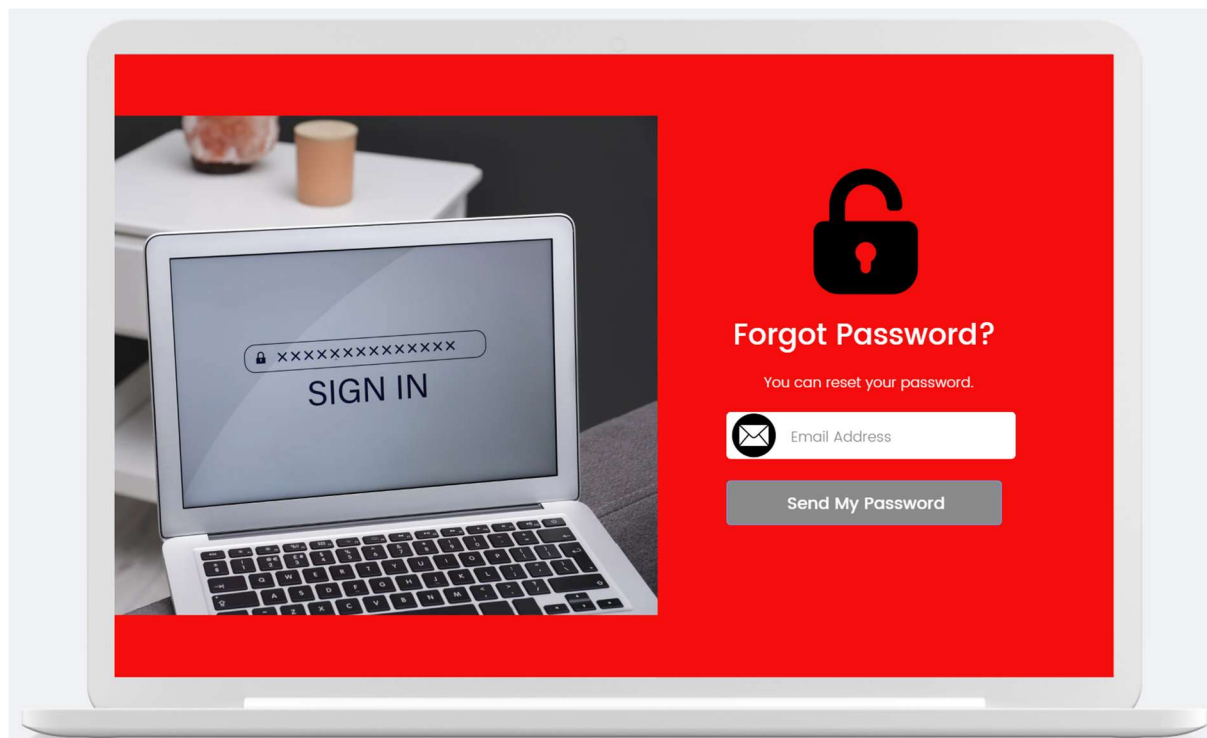
Username

Password

Confirm Password

☐ [Agree to User Terms and Conditions](#)

[Figure 4: Forgot Password]





Forgot Password?

You can reset your password.



4. Security Design

Security is embedded at every level of the WAV2T platform. The system currently uses Django Allauth for authentication, and role-based access control is enforced through Django's permission groups and middleware.

Key security features include:

- **Authentication:** Users log in securely via Django's built-in auth system. All passwords are hashed using PBKDF2.
- **Authorization:** Role-based access ensures only authorized users can access exams, results, or admin functions.
- **Data Encryption:** Sensitive data is encrypted in transit using HTTPS and hashed at rest.
- **Session Management:** Sessions expire after inactivity, and login attempt limits help mitigate brute-force attacks (Django-Axes).
- **Audit Logging:** Administrative actions and login attempts are logged for future monitoring and compliance.

Planned enhancements include multi-factor authentication (MFA), email-based alerts, and SSO integration for production environments.

V. System Implementation

The implementation phase of the WAV2T platform focused on building a secure, modular, and user-centric web application using industry best practices. This section outlines the methodology employed, the organization of the project codebase, implementation strategies, and the testing processes used to validate functionality and performance.

1. Development Methodology

The WAV2T project adopted an Agile development methodology to facilitate continuous improvement, flexibility, and rapid iteration. Development was organized into sprints, with regular check-ins and milestone reviews. Key elements of the methodology included:

- **Iterative Development:** Functional components such as user authentication, exams, and study materials were developed in phases, allowing for early testing and stakeholder feedback.
- **Collaboration:** The team used version control (Git) and GitHub for source code collaboration, ensuring transparency and coordination between developers.